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Module 03 - Network Scanning

Learning Objectives

- Understand network protocol fundamentals (TCP/UDP, OSI/TCP-IP models)
- Master network discovery and host enumeration techniques
- Learn comprehensive port scanning methodologies
- Develop skills in service version detection and OS fingerprinting
- Understand firewall and IDS evasion techniques
- Explore advanced scanning tools and automation strategies
- Master Nmap Scripting Engine (NSE) for vulnerability detection

Network Protocol Fundamentals

TCP vs UDP Comparison

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are core transport layer protocols that define how data packets are transmitted across networks.

Characteristic	TCP Connection-oriented	UDP
Connection Type	protocol	Connectionless protocol
Reliability	Provides error checking and correction	No error checking mechanism
Data Delivery	Guarantees delivery of data	No guarantees of data delivery
Transmission Speed	Slower due to overhead	Faster transmission
Packet Routing	All packets follow the same path	Packets can follow any path
Retransmission	Automatic retransmission possible	No retransmission capability
Use Cases	Web browsing, email, file transfer	Streaming, gaming, DNS queries

TCP Control Flags

Core TCP Flags

- SYN (Synchronize): Initiates three-way handshake between hosts
- ACK (Acknowledgment): Acknowledges successful packet receipt
- FIN (Finished): Indicates no more data from sender
- RST (Reset): Immediately terminates connection
- PSH (Push): Forces immediate processing of buffered data
- URG (Urgent): Indicates urgent data that should be processed immediately

TCP Three-Way Handshake

```
┌─────────┐                    ┌─────────┐
│ Client  │                    │ Server  │
└─────────┘                    └─────────┘
     │                              │
     │ 1. SYN (seq=x)               │
     │ ──────────────────────────►  │
     │                              │
     │ 2. SYN-ACK (seq=y, ack=x+1)  │
     │ ◄──────────────────────────  │
     │                              │
     │ 3. ACK (seq=x+1, ack=y+1)    │
     │ ──────────────────────────►  │
     │                              │
     │    [Connection Established]  │
```

Security Implications:

- SYN flood attacks: Overwhelming server with SYN requests
- Connection hijacking: Exploiting sequence number prediction
- Port scanning detection: Analyzing handshake responses

Network Models and Architecture

OSI (Open Systems Interconnection) Model

Layer	Name	Description	Example Protocols
7	Application	Human-computer interaction interface	HTTP, HTTPS, FTP, SMTP, DNS

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Layer	Name	Description	Example Protocols
6	Presentation	Data formatting, encryption, compression	TLS/SSL, JPEG, PNG, ASCII
5	Session	Session management and control	NetBIOS, RPC, SQL sessions
4	Transport	Reliable data transmission	TCP, UDP, SCTP IP, ICMP,
3	Network	Routing and logical addressing	OSPF, BGP Ethernet, Wi-Fi, ARP
2	Data Link	Node-to-node data transfer	Cables, radio waves, fiber optic
1	Physical	Physical transmission medium	

TCP/IP Model

Layer	Name	Description	Example Protocols
4	Application	Application-level protocols	HTTP, HTTPS, FTP, SSH, Telnet
3	Transport	Host-to-host communication	TCP, UDP
2	Internet	Internetworking and routing	IP, ICMP, ARP
1	Link	Network interface and hardware	Ethernet, Wi-Fi, PPP

Security Considerations:

- Layer-specific attacks: Each layer presents unique vulnerabilities
- Protocol analysis: Understanding protocols aids in vulnerability assessment
- Defense in depth: Security controls at multiple layers

Network Scanning Fundamentals

Core Scanning Objectives

🎯 Primary Goals

- Live host discovery: Identify active systems on the network
- Port enumeration: Discover open ports and running services
- Service identification: Determine service versions and configurations
- Operating system detection: Fingerprint target system architecture
- Security posture assessment: Identify firewalls, IDS/IPS systems

📊 Scanning Methodology

- Reconnaissance: Gather target information and network topology
- Discovery: Identify live hosts and network ranges
- Enumeration: Discover ports, services, and system details
- Vulnerability assessment: Identify potential security weaknesses
- Documentation: Record findings for analysis and reporting

Host Discovery Techniques

🔍 Network Discovery Methods

ARP-Based Discovery (Local Network)

```
# ARP scanningforlocalsubnetdiscovery
arp-scan --local                          #Scan local network segment
arp-scan 192.168.1.0/24                   #Specific subnet ARP scan
arp-scan -I eth0 192.168.1.0/24           #Interface-specific scan
netdiscover -r 192.168.1.0/24             #Active network discovery
netdiscover -r 192.168.1.0/24-P           #Passive network discovery
```

Advantages: Fast, reliable for local networks, bypasses IP-level filtering Limitations: Only works on local subnet, may trigger ARP monitoring

ICMP-Based Discovery

```
# ICMP ping sweeps nmap -sn
192.168.1.0/24 nmap -sn -PE               #No port scan ping sweep
192.168.1.0/24 nmap -sn -PP               #ICMP Echo ping
192.168.1.0/24 nmap -sn -PM               #ICMP Timestamp ping
192.168.1.0/24  fping  -a  -g             #ICMP Address Mask ping
192.168.1.0/24                            #Fast ping alternative
```

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Advantages: Quick host identification, standard network diagnostic Limitations: Often blocked by firewalls, may not reflect actual host status

TCP/UDP-Based Discovery

```
# TCP SYN discovery
nmap -sn -PS22,80,443 192.168.1.0/24       #TCP SYN ping to specific ports
nmap -sn -PA80,443 192.168.1.0/24          #TCP ACK ping

# UDP discovery
nmap -sn -PU53,67,123 192.168.1.0/24       #UDP ping to specific ports
```

Advantages: Bypasses ICMP filtering, targets specific services Limitations: May trigger security alerts, slower than ICMP

Port Scanning Techniques

🔍 Nmap Port Scan Types

TCP Connect Scanning

```
# TCP Connect scans (complete three-way handshake)
nmap -sT target                             # Basic TCP connect scan
nmap -sT -p 1-65535 target                 # Full port range scan
nmap -sT -p- target                        # All 65535 ports
nmap -sT --top-ports 1000 target           # Top 1000 most common ports
```

Characteristics:

- Stealth level: Low (creates full connections)
- Accuracy: High (definitive open/closed status)
- Speed: Slower due to full handshake
- Detection: Easily logged and detected

SYN Stealth Scanning

```
# SYN stealthscans(half-openscanning)
nmap -sS target                             #Basic SYN scan
nmap -sS -T4 target                        #Faster timing template
nmap -sS -p 80,443,22,21,25target          #Specific port list
nmap -sS --scan-delay 1s target            #Custom delay between probes
```

Characteristics:

- Stealth level: Higher (no full connection established)
- Accuracy: High for determining port state
- Speed: Fast (no connection teardown)
- Detection: More difficult to detect than TCP connect

Advanced Scan Types

```
#   Specializedscantechniques
nmap  -sA  target  nmap  -sW              #ACK  scan  (firewall  detection)
target nmap -sM target nmap -             #Window  scan  (OS  detection)
sN  target  nmap  -sF  target             #Maimon  scan  (FIN/ACK)  #Null
nmap -sX target                           scan  (no  flags  set)  #FIN  scan
                                          (FIN  flag  only)  #Xmas  scan
                                          (FIN+PSH+URG flags)
```

Use Cases:

- ACK scans: Firewall rule testing and bypass
- FIN/NULL/Xmas scans: Evasion of simple packet filters
- Window scans: Operating system fingerprinting

📊 Port State Interpretation

Nmap Port States

- Open: Service actively listening and accepting connections
- Closed: Port accessible but no service listening
- Filtered: Packet filtered by firewall, no response received
- Unfiltered: Port accessible but unknown if open or closed
- Open|Filtered: Cannot determine if open or filtered
- Closed|Filtered: Cannot determine if closed or filtered

Service Detection and Version Enumeration

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🔍 Service Version Detection

Basic Service Enumeration

```
# Service versiondetection nmap -sV
target nmap -sV --version- #Basic service version scan
intensity0target nmap -sV --version- #Light version detection
intensity9target nmap -sV --version- #Aggressive version detection
all target #Try all probes
```

Advanced Service Analysis

```
# Comprehensiveserviceanalysis
nmap -A target #Aggressive scan (OS+Version+Scripts+Traceroute)
nmap -sC -sVtarget #Default scripts with version detection
nmap -O target #Operating system detection #OSdetection
nmap -sV -O --osscan-guesstarget with guessing
```

🎯 Banner Grabbing Techniques

Manual Banner Grabbing

```
# Direct serviceinteraction
nc -nv target 21 nc -nv #FTP banner grabbing
target 22 nc -nv target 25 #SSH version information
nc -nv target 80 telnet #SMTP service banner
target 110 #HTTP server information
#POP3 service banner

# HTTP-specific banner grabbing
curl -I http://target # HTTP headers only
wget --server-response --spider http://target # Server response headers
```

Automated Banner Grabbing

```
# Nmap banner grabbing scripts
nmap --script banner target #Generic banner grabbing
nmap --script http-headerstarget #HTTP header enumeration
nmap --script smtp-commandstarget #SMTP command enumeration
nmap --script ssh2-enum-algostarget #SSH algorithm enumeration
```

Operating System Detection

💻 OS Fingerprinting Techniques

Active OS Fingerprinting

```
# Nmap OS detection nmap -O
target nmap -O --osscan- #Basic OS detection
limittarget nmap -O --osscan- #Limit to promising targets
guesstarget nmap -O --max-os- #Aggressive OS guessing
tries2target #Limit OS detection attempts
```

OS Detection Methods

- TCP ISN analysis: Initial sequence number patterns
- TCP options analysis: Window size, option order patterns
- ICMP analysis: Response patterns and error handling
- UDP analysis: Closed port response behavior

Passive OS Fingerprinting

```
# p0f passiveOSfingerprinting
p0f -i eth0-p #Passive analysis of traffic
p0f -f /etc/p0f/p0f.fp-rcapture.pcap #Analysis of packet capture

# Alternative passive tools
ettercap -T -M arp:remote target #Passive scanning with ettercap
```

Advantages: Completely passive, no packets sent to target Limitations: Requires network traffic analysis, less detailed information

UDP Scanning Methodology

📡 UDP Service Discovery

Comprehensive UDP Scanning

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```
# UDP port scanning
nmap -sU target                          # Top 1000 UDP ports
nmap -sU --top-ports 100 target          # Top 100 UDP ports
nmap -sU -p 53,67,68,69,123,161,162 target # Common UDP services
nmap -sU -sV target                      # UDP scan with version detection
```

Common UDP Services

- Port 53: DNS (Domain Name System)
- Port 67/68: DHCP (Dynamic Host Configuration Protocol)
- Port 69: TFTP (Trivial File Transfer Protocol)
- Port 123: NTP (Network Time Protocol)
- Port 161/162: SNMP (Simple Network Management Protocol)
- Port 500: IKE (Internet Key Exchange)
- Port 514: Syslog
- Port 1434: MS SQL Server Browser

UDP Scanning Challenges

- Reliability issues: UDP is connectionless, responses not guaranteed
- False positives: Firewalls may drop packets silently
- Speed limitations: Slower than TCP scanning due to rate limiting

Nmap Scripting Engine (NSE)

🔧 NSE Categories and Usage

Vulnerability Detection Scripts

```
# Comprehensive vulnerabilityscanning
nmap --script vuln target                #All vulnerability scripts
nmap --script vuln,safetarget            #Safe vulnerability scripts only
nmap --script smb-vuln-*target           #SMB-specific vulnerabilities
nmap --script http-vuln-*target          #HTTP vulnerability scripts
nmap --script ssl-* target               #SSL/TLS security scripts
```

Service Enumeration Scripts

```
#  Service-specific  enumeration
nmap  --script  smb-enum-*target         #SMB   enumeration   scripts
nmap --script dns-* target nmap          #DNS   enumeration   scripts
--script  http-enum  target  nmap        #HTTP  directory  enumeration
--script  ftp-*  target  nmap  --        #FTP  service  scripts  #SSH
script ssh-* target                      enumeration scripts
```

Brute Force Scripts

```
#  Authentication  brute  forcing  nmap  --script  ssh-brute  --script-args
userdb=users.txt,passdb=pass.txt  target  nmap  --script  ftp-brute  --script-args
userdb=users.txt,passdb=pass.txt  target  nmap  --script  http-brute  --script-args
userdb=users.txt,passdb=pass.txt  target  nmap  --script  smb-brute  --script-args
userdb=users.txt,passdb=pass.txt target
```

Custom Script Parameters

```
# Script argument customization
nmap --script http-enum --script-args http-enum.basepath='/admin/' target
nmap --script smb-enum-shares --script-args smbdomain=WORKGROUP target
nmap --script ssl-cert --script-args ssl-cert.timeout=5s target
```

Advanced Scanning Techniques

Stealth and Evasion Methods

Timing and Performance Tuning

```
# Timing templates (0-5, slowest to fastest)
nmap  -T0  target                        # Paranoid timing (5+ minutes between probes)
nmap  -T1  target                        # Sneaky timing (15 seconds between probes)
nmap  -T2  target                        # Polite timing (0.4 seconds between probes)
nmap  -T3  target                        # Normal timing (default)
nmap  -T4  target                        # Aggressive timing
nmap -T5 target                          # Insane timing (fastest)
```

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```
# Custom timing controls
nmap --min-rate 100 target              #Minimum packet rate
nmap --max-rate 1000 target             #Maximum packet rate
nmap --scan-delay 2s target             #Delay between probes
nmap --max-retries 2 target             #Maximum probe retries
```

Packet Fragmentation

```
# Fragmentationtechniques
nmap -f target                          #Fragment packets into 8-byte chunks
nmap -ff target                         #Fragment into 16-byte chunks
nmap --mtu 24target                     #Custom MTU size (multiple of 8)
nmap --data-length25target              #Append random data to packets
```

Source Address Manipulation

```
# Decoy scanning
nmap -D RND:10 target                   #10random decoy addresses
nmap -D decoy1,decoy2,decoy3target      #Specific decoy addresses
nmap -D RND:5,ME target                 #5decoys plus real address

# Source port manipulation
nmap -g 53 target nmap -g 88            #Source port 53 (DNS)
target nmap --source-                   #Source port 88 (Kerberos)
port53target                            #Alternative syntax

# IDLE/Zombie scanning
nmap -sI zombie_host target             #IDLE scan using zombie host
```

Advanced Evasion Techniques

```
# Firewall and IDS evasion
nmap -f -D RND:10 --randomize-hosts target # Multiple evasion techniques
nmap --spoof-mac 0 target               # Random MAC address spoofing
nmap --spoof-mac Dell target            # Vendor-specific MAC spoofing
nmap --ip-options "L 192.168.1.1" target # Loose source routing
nmap --ttl 64 target                    # Custom TTL value
```

High-Speed Scanning with Masscan

⚡ Masscan Configuration and Usage

Basic Masscan Operations

```
# High-speed port scanning
masscan -p1-65535 target --rate=1000        # Full port range at 1000 pps
masscan -p80,443,8080,8443 target --rate=10000 # Specific ports at high speed
masscan --top-ports 100 target --rate=5000 # Top 100 ports scan

# Output formatting
masscan -p1-65535 target --rate=1000 -oG masscan.out # Grepable output
masscan -p1-65535 target --rate=1000 -oJ masscan.json # JSON output
masscan -p1-65535 target --rate=1000 -oX masscan.xml # XML output
```

Masscan with Nmap Integration

```
# Two-stage scanning approach
# Stage 1: Fast port discovery with masscan
masscan -p1-65535 target --rate=5000 -oG discovered_ports.txt

# Stage 2: Detailed analysis with nmap
nmap -sV -sC -iL discovered_ports.txt -p $(cat discovered_ports.txt | grep "Host:" | cut -d' ' -f5 | cut -d'/' -f1 | sort -u | tr '\n' ',' | sed
's/,$//')
```

🌐 Large-Scale Network Scanning

Internet-Wide Scanning Considerations

```
# Responsible internet scanning masscan -p80,443 0.0.0.0/0 --
rate=1000 --excludefile exclude.txt # Note: Always include
exclude files for critical infrastructure

# Network range scanning
masscan -p22,80,443 10.0.0.0/8 --rate=10000 --randomize-hosts
masscan -p1-1000 192.168.0.0/16 --rate=5000 --wait=5
```

Ethical Considerations:

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- Rate limiting: Avoid overwhelming target networks
- Exclude lists: Respect network operator preferences
- Legal compliance: Ensure proper authorization for scanning
- Responsible disclosure: Report critical vulnerabilities appropriately

IPv6 Scanning Strategies

🌐 IPv6 Network Discovery

IPv6 Address Discovery

```
# IPv6 host discovery
nmap -6 target_ipv6                    #Basic IPv6 scan #IPv6 ping
nmap -6 -sn2001:db8::/32               sweep   #IPv6   alive   host
alive6 eth0                            detection

# IPv6 address enumeration
atk6-address6 2001:db8::1              #Generate potential addresses
atk6-detect-new-ip6 eth0               #Detect new IPv6 addresses
```

IPv6 Scanning Challenges

- Address space size: 128-bit addresses create enormous search space
- Address allocation patterns: Understanding common IPv6 addressing schemes
- Reduced toolset: Fewer tools support comprehensive IPv6 scanning
- Security assumptions: IPv6 networks often have weaker security controls

Automation and Scripting

🤖 Automated Scanning Workflows

Comprehensive Scanning Script

```
#!/bin/bash
# network_scanner.sh - Automated network reconnaissance

TARGET=$1
OUTPUT_DIR="scan_$(date +%Y%m%d_%H%M%S)"
THREADS=50

if [ -z "$TARGET" ]; then
    echo "Usage: $0 <target_network>"
    exit 1
fi

mkdir -p "$OUTPUT_DIR"
echo "[*] Starting comprehensive network scan of $TARGET"
echo "[*] Results will be saved in $OUTPUT_DIR"

# Phase 1: Host Discovery
echo "[+] Phase 1: Host Discovery"
nmap -sn "$TARGET" | grep "Nmap scan report" | awk '{print $5}' > "$OUTPUT_DIR/live_hosts.txt"
host_count=$(wc -l < "$OUTPUT_DIR/live_hosts.txt")
echo "[+] Discovered $host_count live hosts"

# Phase 2: Port Scanning
echo "[+] Phase 2: Port Scanning"
nmap -sS -T4 -iL "$OUTPUT_DIR/live_hosts.txt" -oN "$OUTPUT_DIR/port_scan.nmap" \
     -oG "$OUTPUT_DIR/port_scan.gnmap" -oX "$OUTPUT_DIR/port_scan.xml"

# Phase 3: Service Detection
echo "[+] Phase 3: Service Detection"
nmap -sV -sC -iL "$OUTPUT_DIR/live_hosts.txt" -oN "$OUTPUT_DIR/service_scan.nmap" \
     -oX "$OUTPUT_DIR/service_scan.xml"

# Phase 4: Vulnerability Assessment
echo "[+] Phase 4: Vulnerability Assessment"
nmap --script vuln -iL "$OUTPUT_DIR/live_hosts.txt" -oN "$OUTPUT_DIR/vuln_scan.nmap"

# Phase 5: OS Detection
echo "[+] Phase 5: OS Detection"
nmap -O --osscan-guess -iL "$OUTPUT_DIR/live_hosts.txt" -oN "$OUTPUT_DIR/os_detection.nmap"

echo "[+] Scan completed. Results available in $OUTPUT_DIR"
```

Parallel Scanning with GNU Parallel

```
# Parallel host scanning
parallel -j 10 nmap -sS -T4 {} :::: live_hosts.txt
```

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```
# Parallel service detection
parallel -j 5 nmap -sV -p {} target ::: $(cat open_ports.txt)
```

Network Scanning Defense and Detection

🛡️ Defensive Measures

Scan Detection Techniques

- **Connection monitoring**: Tracking rapid connection attempts
- **Port knock detection**: Identifying systematic port probing
- **Timing analysis**: Detecting non-human scanning patterns
- **Honeypots**: Deploying decoy services to detect scanning

Countermeasures Implementation

```
# Iptables rate limiting
iptables -A INPUT -p tcp --dport 22 -m limit --limit 3/min -j ACCEPT
iptables -A INPUT -p tcp --dport 22 -j DROP

# Port knocking implementation
iptables -A INPUT -p tcp --dport 22 -m recent --update --seconds 60 --name SSH --rsource -j DROP
iptables -A INPUT -p tcp --dport 22 -m recent --set --name SSH --rsource -j ACCEPT
```

Additional Resources

📚 Essential Reading

- Nmap Official Documentation - Comprehensive Nmap reference guide
- NIST SP 800-115 - Network Security Testing guidelines
- RFC 793 - TCP Protocol - TCP specification and behavior

🔬 Research Papers and Case Studies

- "Network Scanning Techniques and Defense" - Academic analysis of scanning methods
- SANS Network Security Monitoring - Detection and response strategies
- IPv6 Security Research - IPv6-specific security considerations

🛠️ Professional Tools and Resources

- Masscan Documentation - High-speed port scanner
- **Nmap Scripting Engine**: Custom script development
- **Zmap Project**: Internet-wide scanning tools
- **NMAP Front-ends**: Zenmap, NmapFE, WebMap
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